ST10490015

PROG6212_Part 1

RaceDay_API_Endpoint_Plan

RaceDay API Endpoint Plan

Part 2 lists every endpoint that the RaceDay RESTful API will be offering, in this plan. It is a straight copy of the six entities of the ERD in RaceDay_ERD.pdf and includes Authentication, User Profile, Events, Routes, Categories, Event Enrolments and Results.

Role Required legend

- None – public, no login required.
- Any – Any User Logged In (Role).
- The Organiser role is limited to a logged in user. Only a logged-in user with the Organiser role can be an Organiser.
- Participant – logged in user with only the Participant role.

All routes which contain {id} and require Organiser will also ensure that the Organiser logged in is the owner of the resource (e.g. OrganiserID to that event) or the route is failing, returning 403 Forbidden.

All Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
1. Authentication					
POST	/api/auth/register	Registers a new RaceDay account as either an Organiser or a Participant.	None	{ fullName, email, password, role, phoneNumber, dateOfBirth }	201 Created – new user id and role 400 Bad Request – validation failed 409 Conflict – email already registered
POST	/api/auth/login	Authenticates a registered user and issues a JWT access token that encodes their role.	None	{ email, password }	200 OK – JWT token and user role 401 Unauthorized – invalid email or password
2. User Profile					
GET	/api/users/me	Returns the profile of the currently logged-in user.	Any	None	200 OK – user profile 401 Unauthorized

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
PUT	/api/users/me	Updates the logged-in user's own profile details.	Any	{ fullName, phoneNumber, dateOfBirth }	200 OK – updated profile 400 Bad Request 401 Unauthorized
3. Events					
GET	/api/events	Returns a list of all RaceDay events.	None	None	200 OK – list of events
GET	/api/events/{id}	Returns full details of a single event by its ID.	None	None	200 OK – event details 404 Not Found
POST	/api/events	Creates a new event owned by the logged-in Organiser.	Organiser	{ eventName, description, eventDate, location, eventType }	201 Created – new event 400 Bad Request 401 Unauthorized
PUT	/api/events/{id}	Updates an event; only the Organiser who owns it may edit it.	Organiser	{ eventName, description, eventDate, location, eventType }	200 OK – updated event 400 Bad Request 403 Forbidden – not the owner 404 Not Found
DELETE	/api/events/{id}	Deletes an event owned by the logged-in Organiser.	Organiser	None	204 No Content 403 Forbidden 404 Not Found
4. Routes					
GET	/api/events/{id}/routes	Returns the course route(s) for an event so participants can prepare for race day.	None	None	200 OK – list of routes 404 Not Found
POST	/api/events/{id}/routes	Adds a new route to an event.	Organiser	{ routeName, distanceKM, startPoint, endPoint, elevationGainM, mapURL }	201 Created – new route 400 Bad Request 403 Forbidden 404 Not Found
5. Categories					
GET	/api/events/{id}/categories	Returns all entry categories available for a specific event.	None	None	200 OK – list of categories 404 Not Found
POST	/api/events/{id}/categories	Creates a new category for an event.	Organiser	{ categoryName, distanceKM, minAge, maxAge, entryFee }	201 Created – new category 400 Bad Request 403 Forbidden 404 Not Found

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
PUT	/api/categories/{id}	Updates the details of an existing category.	Organiser	{ categoryName, distanceKM, minAge, maxAge, entryFee }	200 OK – updated category 400 Bad Request 403 Forbidden 404 Not Found
DELETE	/api/categories/{id}	Deletes a category. Blocked if participants are already enrolled in it.	Organiser	None	204 No Content 403 Forbidden 404 Not Found 409 Conflict – existing enrolments
6. Event Enrolments					
POST	/api/events/{id}/enrolments	Enrols the logged-in Participant into an event under a chosen category.	Participant	{ categoryId }	201 Created – enrolment record 400 Bad Request 401 Unauthorized 404 Not Found – event or category 409 Conflict – already enrolled in this event
GET	/api/users/me/enrolments	Returns the logged-in Participant's own enrolment history.	Participant	None	200 OK – list of enrolments 401 Unauthorized
GET	/api/events/{id}/enrolments	Returns all enrolments for an event, for the Organiser who owns it.	Organiser	None	200 OK – list of enrolments 403 Forbidden – not the owner 404 Not Found
DELETE	/api/enrolments/{id}	Cancels the logged-in Participant's own enrolment.	Participant	None	204 No Content 403 Forbidden – not the owner 404 Not Found 409 Conflict – event already started
7. Results					
POST	/api/enrolments/{id}/result	Captures the finish time and position for a participant's enrolment.	Organiser	{ finishTime, finishPosition, categoryPosition, status }	201 Created – result recorded 400 Bad Request 403 Forbidden – not the event owner 404 Not Found 409 Conflict – result already captured
PUT	/api/results/{id}	Corrects a previously captured result.	Organiser	{ finishTime, finishPosition, categoryPosition, status }	200 OK – updated result 400 Bad Request 403 Forbidden 404 Not Found
GET	/api/events/{id}/results	Returns the public results/leaderboard for an event.	None	None	200 OK – list of results 404 Not Found
GET	/api/users/me/results	Returns the logged-in Participant's own performance history across all events.	Participant	None	200 OK – list of results 401 Unauthorized